

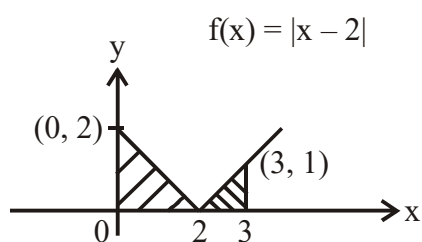
61. Let $f(x) = |x - 2|$ and $g(x) = f(f(x))$, $x \in [0, 4]$. Then $\int_0^3 (g(x) - f(x)) dx$ is equal to :

यदि $x \in [0, 4]$ के लिए $f(x) = |x - 2|$ तथा $g(x) = f(f(x))$ हैं, तो $\int_0^3 (g(x) - f(x)) dx$ बराबर है:

- (A) 0 (B) $\frac{1}{2}$ (C) $\frac{3}{2}$ (D) 1

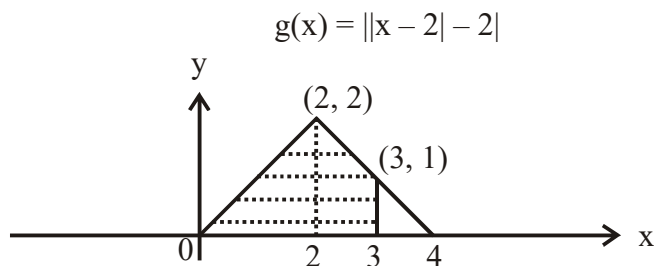
Ans. D

Sol. $f(x) = |x - 2|$ $g(x) = f(|x - 2|) = ||x - 2| - 2|$



$$\int_0^3 f(x) dx = \text{Area of shaded region}$$

$$= \frac{1}{2} \times 2 \times 2 + \frac{1}{2} \times 1 \times 1 = \frac{5}{2}$$



$$\int_0^3 g(x) dx = \text{Area of shaded region}$$

$$= \frac{1}{2} \times 2 \times 2 + \frac{1}{2} (1) \cdot (1 + 2) = \frac{7}{2}$$

$$\therefore \int_0^3 (g(x) - f(x)) dx = \int_0^3 g(x) dx - \int_0^3 f(x) dx = \frac{7}{2} - \frac{5}{2} = 1$$

62. For $a > 0$, let the curves $C_1 : y^2 = ax$ and $C_2 : x^2 = ay$ intersect at origin O and a point P. Let the line $x = b$ (0

$b < a$) intersect the chord OP and the x-axis at points Q and R, respectively. If the line $x = b$ bisects the area bounded by the curves, C_1 and C_2 , and the area of $\Delta OQR = \frac{1}{2}$, 'a' satisfies the equation :

$a > 0$ के लिए, माना वक्र $C_1 : y^2 = ax$ तथा $C_2 : x^2 = ay$, मूलबिन्दु O तथा एक बिन्दु P पर काटते हैं। माना रेखा $x = b$ ($0 < b < a$) , जीवा OP तथा x-अक्ष को क्रमशः बिन्दुओं Q तथा R पर काटती है। यदि रेखा $x = b$, वक्रों C_1 तथा C_2 द्वारा परिबद्ध क्षेत्र को समद्विभाजित करती है तथा ΔOQR का क्षेत्रफल $= \frac{1}{2}$ है, तो 'a' जिस समीकरण को संतुष्ट करता है, वह है :

(A) $x^6 + 6x^3 - 4 = 0$

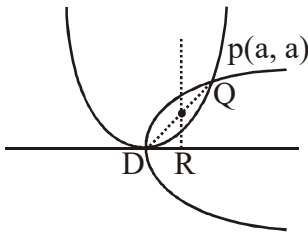
(B) $x^6 - 12x^3 - 4 = 0$

(C) $x^6 - 6x^3 + 4 = 0$

(D) $x^6 - 12x^3 + 4 = 0$

Ans. D

Sol.



$$\text{Total area} = \frac{16ab}{3} = \frac{16 \times \frac{a}{4} \times \frac{a}{4}}{3} = \frac{a^2}{3}$$

$$\int_0^b \left(\sqrt{ax} - \frac{x^2}{a} \right) dx = \frac{a^2}{6}$$

$$\frac{1}{2} b^2 = \frac{1}{2}$$

$$b = 1$$

$$\frac{2}{3} \sqrt{a} b^{3/2} - \frac{b^3}{3a} = \frac{a^2}{6}$$

$$\dots\dots\dots(1)$$

Put in equation (1)

$$a^6 - 12a^3 + 4 = 0$$

63. If a curve $y = f(x)$ passes through the point (1, 2) and satisfies $x \frac{dy}{dx} + y = bx^4$, then for what value of b,

$$\int_1^2 f(x) dx = \frac{62}{5}?$$

यदि एक वक्र $y = f(x)$ बिन्दु (1, 2) से होकर जाता है तथा $x \frac{dy}{dx} + y = bx^4$ को संतुष्ट करता है, तो b के किस मान

के लिए $\int_1^2 f(x) dx = \frac{62}{5}$ है ?

- (A) 10 (B) 5 (C) $\frac{31}{5}$ (D) $\frac{62}{5}$

Ans. A

Sol. $\frac{xdy}{dx} + y = bx^4$

$$\int d(xy) = \int bx^4 dx$$

$$xy = \frac{bx^5}{5} + c$$

$$y = \frac{bx^4}{5} + \frac{c}{x}$$

$$\text{Passes through } (1, 2) \Rightarrow \frac{b}{5} + c = 2 \quad \dots(1)$$

$$\int_1^2 f(x) dx = \frac{62}{5}$$

$$\left(\frac{bx^5}{25} + c \ln x \right)_1^2 = \frac{62}{5}$$

$$\frac{31b}{25} + c \ln 2 = \frac{62}{5} \quad \dots(2)$$

$\Rightarrow$ solving equation (1) and (2)

$$\Rightarrow b = 10 \text{ and } c = 0$$

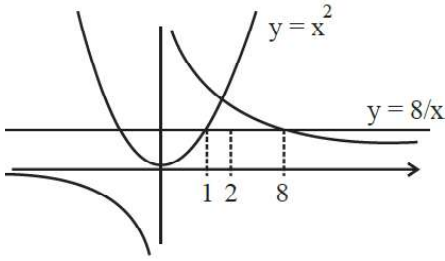
64. The area of the region given by $\{(x, y) : xy \leq 8, 1 \leq y \leq x^2\}$ is :

$\{(x, y) : xy \leq 8, 1 \leq y \leq x^2\}$ द्वारा दिए क्षेत्र का क्षेत्रफल है :

- (A) $8 \log_e 2 + \frac{7}{6}$ (B) $16 \log_e 2 - \frac{14}{3}$ (C) $8 \log_e 2 - \frac{13}{3}$ (D) $16 \log_e 2 + \frac{7}{3}$

Ans. B

Sol. $xy \leq 8$ $1 \leq y \leq x^2$



$$\begin{aligned} \text{Area} &= \int_1^2 (x^2 - 1) dx + \int_2^8 \left(\frac{8}{x} - 1 \right) dx \\ &= \left(\frac{x^3}{3} - x \right)_1^2 + 8(\ln x)_2^8 - (x)_2^8 \\ &= \frac{7}{3} + 8(2\ln 2) - 7 \\ &= 16\ln 2 - \frac{14}{3} \end{aligned}$$

65. The value of the integral $\int \frac{\sin \theta \cdot \sin 2\theta (\sin^6 \theta + \sin^4 \theta + \sin^2 \theta) \sqrt{2\sin^4 \theta + 3\sin^2 \theta + 6}}{1 - \cos 2\theta} d\theta$ is :

(where c is a constant of integration)

समाकलन $\int \frac{\sin \theta \cdot \sin 2\theta (\sin^6 \theta + \sin^4 \theta + \sin^2 \theta) \sqrt{2\sin^4 \theta + 3\sin^2 \theta + 6}}{1 - \cos 2\theta} d\theta$ बराबर हैं :

(जहाँ c एक समाकलन अचर है)

- (A) $\frac{1}{18} [11 - 18\sin^2 \theta + 9\sin^4 \theta - 2\sin^6 \theta]^{\frac{3}{2}} + c$
- (B) $\frac{1}{18} [9 - 2\cos^6 \theta - 3\cos^4 \theta - 6\cos^2 \theta]^{\frac{3}{2}} + c$
- (C) $\frac{1}{18} [11 - 18\cos^2 \theta + 9\cos^4 \theta - 2\cos^6 \theta]^{\frac{3}{2}} + c$
- (D) $\frac{1}{18} [9 - 2\sin^6 \theta - 3\sin^4 \theta - 6\sin^2 \theta]^{\frac{3}{2}} + c$

Ans. C

Sol.
$$I = \int \frac{\sin \theta \cdot \sin 2\theta (\sin^6 \theta + \sin^4 \theta + \sin^2 \theta) \sqrt{2 \sin^4 \theta + 3 \sin^2 \theta + 6}}{1 - \cos 2\theta} d\theta$$

$$I = \int \frac{\sin \theta (2 \sin \theta \cos \theta) (\sin^6 \theta + \sin^4 \theta + \sin^2 \theta) \sqrt{2 \sin^4 \theta + 3 \sin^2 \theta + 6}}{2 \sin^2 \theta} d\theta$$

$$I = \int (\sin^6 \theta + \sin^4 \theta \sin^2 \theta) \cos \theta \sqrt{2 \sin^4 \theta + 3 \sin^2 \theta + 6} d\theta$$

$$I = \int (\sin^5 \theta + \sin^3 \theta \sin \theta) \cos \theta \sqrt{2 \sin^6 \theta + 3 \sin^4 \theta + 6 \sin^2 \theta} d\theta$$

Let, $2 \sin^6 \theta + 3 \sin^4 \theta + 6 \sin^2 \theta = t$

$$12 (\sin^5 \theta + \sin^3 \theta + \sin \theta) \cos \theta d\theta = dt$$

$$(\sin^5 \theta + \sin^3 \theta + \sin \theta) \cos \theta d\theta = \frac{dt}{12}$$

$$I = \frac{1}{12} \int t^{\frac{1}{2}} dt$$

$$= \frac{1}{18} t^{3/2} + C$$

$$= \frac{1}{18} (2 \sin^6 \theta + 3 \sin^4 \theta + 6 \sin^2 \theta)^{\frac{3}{2}} + C$$

$$= \frac{1}{18} (2(1 - \cos^2 \theta)^3 + 3(1 - \cos^2 \theta)^2 + 6(1 - \cos^2 \theta)) + C$$

$$= \frac{1}{18} (11 - 2 \cos^6 \theta + 9 \cos^4 \theta - 18 \cos^2 \theta)^{\frac{3}{2}} + C$$

66. If $\phi(x) = \frac{1}{\sqrt{x}} \int_{\frac{\pi}{4}}^x (4\sqrt{2} \sin t - 3\phi'(t)) dt$, $x > 0$, then $\phi'\left(\frac{\pi}{4}\right)$ is equal to :

यदि $\phi(x) = \frac{1}{\sqrt{x}} \int_{\frac{\pi}{4}}^x (4\sqrt{2} \sin t - 3\phi'(t)) dt$, $x > 0$ है, तो $\phi'\left(\frac{\pi}{4}\right)$ बराबर है :

(A) $\frac{8}{6 + \sqrt{\pi}}$

(B) $\frac{4}{6 + \sqrt{\pi}}$

(C) $\frac{8}{\sqrt{\pi}}$

(D) $\frac{4}{6 - \sqrt{\pi}}$

Ans. A

Sol. $\phi'(x) = \frac{1}{\sqrt{x}} \left[(4\sqrt{2} \sin x - 3\phi'(x)) \cdot 1 - 0 \right] - \frac{1}{2} x^{-3/2}$

$$\int_{\pi/4}^x (4\sqrt{2} \sin t - 3\phi'(t)) dt$$

$$\phi'\left(\frac{\pi}{4}\right) = \frac{2}{\sqrt{\pi}} \left[4 - 3\phi'\left(\frac{\pi}{4}\right) \right] + 0$$

$$\left(1 + \frac{6}{\sqrt{\pi}}\right) \phi'\left(\frac{\pi}{4}\right) = \frac{8}{\sqrt{\pi}}$$

$$\phi'\left(\frac{\pi}{4}\right) = \frac{8}{\sqrt{\pi} + 6}$$

67. $\int_0^{1/2} \frac{x^2}{(1-x^2)^{3/2}} dx$ is $\frac{k}{6}$, then k is equal to :

यदि समाकल $\int_0^{1/2} \frac{x^2}{(1-x^2)^{3/2}} dx$ का मान $\frac{k}{6}$ है, तो k बराबर है :

(A) $2\sqrt{3} - \pi$ (B) $2\sqrt{3} + \pi$ (C) $3\sqrt{2} + \pi$ (D) $3\sqrt{2} - \pi$

Ans. A

Sol. $\frac{k}{6} = \int_0^{1/2} \frac{x^2}{(1-x^2)^{3/2}} dx$ $x = \sin \theta; dx = \cos \theta d\theta$

$$\Rightarrow \frac{k}{6} = \int_0^{\pi/6} \frac{\sin^2 \theta}{(1-\sin^2 \theta)^{3/2}} \cdot \cos \theta d\theta \Rightarrow \frac{k}{6} = \int_0^{\pi/6} \frac{\sin^2 \theta}{\cos^3 \theta} \cdot \cos \theta d\theta$$

$$\Rightarrow \frac{k}{6} = \int_0^{\pi/6} \tan^2 \theta d\theta = \int_0^{\pi/6} (\sec^2 \theta - 1) d\theta \Rightarrow \frac{k}{6} = (\tan \theta - \theta)_0^{\pi/6} = \left(\frac{1}{\sqrt{3}} - \frac{\pi}{6} \right) = \frac{2\sqrt{3} - \pi}{6}$$

$$\Rightarrow k = 2\sqrt{3} - \pi$$

68. Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} = (y+1)\left((y+1)e^{x/2} - x\right)$, $0 < x < 2.1$, with $y(2) = 0$.

Then the value of $\frac{dy}{dx}$ at $x = 1$ is equal to :

माना अवकल समीकरण $\frac{dy}{dx} = (y+1)((y+1)e^{x^2/2} - x)$, $0 < x < 2.1$, $y(2) = 0$ का हल $y = y(x)$ है। तो $x = 1$ पर $\frac{dy}{dx}$ का

मान बराबर है :

- (A) $\frac{5e^{1/2}}{(e^2 + 1)^2}$ (B) $-\frac{2e^2}{(1+e^2)^2}$ (C) $\frac{-e^{3/2}}{(e^2 + 1)^2}$ (D) $\frac{e^{5/2}}{(1 + e^2)^2}$

Ans. C

Sol. $\frac{dy}{dx} = (y+1)((y+1)e^{x^2/2} - x)$

$$y + 1 = Y$$

$$\frac{dy}{dx} = \frac{dY}{dx}$$

$$\frac{dY}{dx} = Y(Ye^{x^2/2} - x)$$

$$\frac{dY}{dx} = Y^2 e^{x^2/2} - xY$$

$$\frac{1}{Y} \frac{dY}{dx} = Ye^{x^2/2} - x$$

$$\frac{1}{Y} = P$$

$$\frac{dP}{dx} = -\frac{1}{Y^2} \frac{dY}{dx}$$

$$\frac{1}{Y^2} \frac{dY}{dx} = e^{x^2/2} - \frac{x}{Y}$$

$$\frac{-dP}{dx} = e^{x^2/2} - Px$$

$$\frac{dP}{dx} = -e^{x^2/2} + Px$$

$$\frac{dP}{dx} - Px = e^{x^2/2}$$

$$\text{I.F.} = e^{x^2/2}$$

$$-P = (x + c)e^{x^2/2}$$

$$\frac{-1}{Y} = (x+c)e^{x^2/2}$$

$$\frac{-1}{y+1} = (x+c)e^{x^2/2}$$

$$y+1 = \frac{-1}{(x+c)e^{x^2/2}}$$

$$x=2, y=0$$

$$0+1 = \frac{-1}{(2+c)e^{x^2/2}}$$

$$(2+c)e^{x^2/2} = -1$$

$$2+c = \frac{-1}{e^{x^2/2}}$$

$$c = -2 - \frac{1}{e^{x^2/2}}$$

$$y+1 = \frac{-1}{\left(x-2 - \frac{1}{e^{x^2/2}}\right)e^{x^2/2}}$$

$$\left(\frac{dy}{dx}\right)_{\text{at } x=1} = \frac{-e^{3/2}}{(1+e^2)^2}$$

69. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuously differentiable function such that $f(2) = 6$ and $f'(2) = \frac{1}{48}$. If

$$\int_6^{f(x)} 4t^3 dt = (x-2)g(x), \text{ then } \lim_{x \rightarrow 2} g(x) \text{ is equal to :}$$

माना $f : \mathbb{R} \rightarrow \mathbb{R}$ एक संतततः अवकलनीय (continuously differentiable) फलन इस प्रकार है कि $f(2) = 6$ तथा $f'(2) = \frac{1}{48}$.

$$\text{यदि } \int_6^{f(x)} 4t^3 dt = (x-2)g(x), \text{ तो } \lim_{x \rightarrow 2} g(x) \text{ बराबर है :}$$

(A) 18

(B) 36

(C) 24

(D) 12

Ans. A

sol. $\int_6^{f(x)} 4t^3 dt = (x-2)g(x)$

$4(f(x)). f'(x) = g'(x)(x-2) + g(x)$

put $x = 2$,

$\frac{4(6)^3 \cdot 1}{48} g(2)$

$\lim_{x \rightarrow 2} g(x) = 18$

70. Let $A = \{(x, y) \in \mathbb{R}^2 : y \geq 0, 2x \leq y \leq \sqrt{4 - (x-1)^2} \text{ and}$

$B = \{(x, y) \in \mathbb{R} \times \mathbb{R} : 0 \leq y \leq \min\{2x, \sqrt{4 - (x-1)^2}\}\}.$

Then the ratio of the area of A to the area of B is :

माना $A = \{(x, y) \in \mathbb{R}^2 : y \geq 0, 2x \leq y \leq \sqrt{4 - (x-1)^2} \text{ और}$

$B = \{(x, y) \in \mathbb{R} \times \mathbb{R} : 0 \leq y \leq \min\{2x, \sqrt{4 - (x-1)^2}\}\}$ है।

तब A के क्षेत्रफल तथा B के क्षेत्रफल का अनुपात है :

(A) $\frac{\pi+1}{\pi-1}$

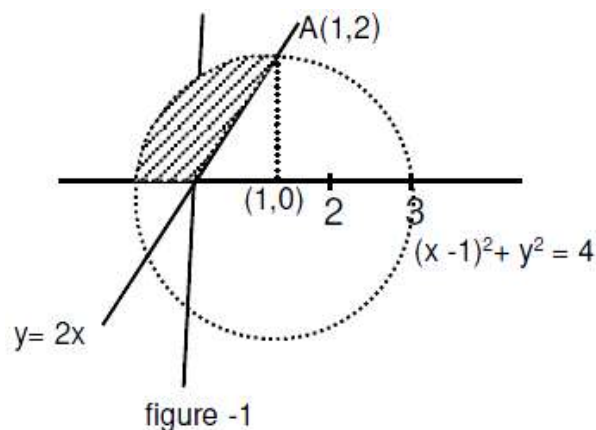
(B) $\frac{\pi}{\pi+1}$

(C) $\frac{\pi}{\pi-1}$

(D) $\frac{\pi-1}{\pi+1}$

Ans. D

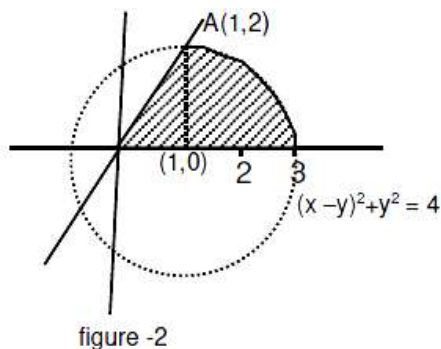
Sol. $2x \leq y \leq \sqrt{4 - (x-1)^2} : y \geq 0$



area of A : area of shaded Region as show in figure-1

$$\therefore \text{area of A} = \frac{\pi \times 4}{2} - \left(\frac{1}{2} \times 1 \times 2 \right)$$

$$= \pi - 1$$



area of B : area of shaded Region in as show in above figure-2

$$\therefore \text{area of B} = \frac{1}{2} \times 1 \times 2 + \frac{\pi \times 4}{4} = (1 + \pi)$$

$$\frac{A}{B} = \frac{\text{area of A}}{\text{area of B}} = \frac{\pi - 1}{\pi + 1}$$

71. If $\int \sin^{-1} \left(\sqrt{\frac{x}{1+x}} \right) dx = A(x) \tan^{-1}(\sqrt{x}) + B(x) + C$, where C is a constant of integration, then the ordered pair (A(x), B(x)) can be :

यदि $\int \sin^{-1} \left(\sqrt{\frac{x}{1+x}} \right) dx = A(x) \tan^{-1}(\sqrt{x}) + B(x) + C$, जहाँ C एक समाकलन अचर है, तो क्रमित युग्म (A(x), B(x)) हो सकता है :

- (A) $(x+1, \sqrt{x})$ (B) $(x-1, \sqrt{x})$ (C) $(x-1, -\sqrt{x})$ (D) $(x+1, -\sqrt{x})$

Ans. D

Sol. $I = \int \sin^{-1} \left(\frac{\sqrt{x}}{\sqrt{1+x}} \right) dx$

$$\int \tan^{-1} \left(\sqrt{x} \right) dx = x \tan^{-1} \sqrt{x} - \int \frac{1}{1+x} \cdot \frac{1}{2\sqrt{x}} \cdot x dx + C = x \tan^{-1} \sqrt{x} - \frac{1}{2} \int \frac{t \cdot 2t \cdot dt}{1+t^2} + C \quad (x = t^2)$$

$$= x \tan^{-1} \sqrt{x} - \int \frac{t^2}{1+t^2} dt + C = x \tan^{-1} \sqrt{x} - t + \tan^{-1} t + C = x \tan^{-1} \sqrt{x} - \sqrt{x} + \tan^{-1} \sqrt{x} + C$$

$$= (x+1) \tan^{-1} \sqrt{x} - \sqrt{x} + c \quad \Rightarrow \quad A(x) = x+1 \Rightarrow B(x) = -\sqrt{x}$$

72. The value of α for which $4\alpha \int_{-1}^2 e^{-\alpha|x|} dx = 5$, is

α का वह मान, जिसके लिए $4\alpha \int_{-1}^2 e^{-\alpha|x|} dx = 5$ है, है :

- (A) $\log_e \sqrt{2}$ (B) $\log_e 2$ (C) $\log_e \left(\frac{4}{3}\right)$ (D) $\log_e \left(\frac{3}{2}\right)$

Ans. B

Sol. $4\alpha \int_{-1}^2 e^{-\alpha|x|} dx = 5$

$$4\alpha \left(\int_{-1}^0 e^{-\alpha|x|} dx + \int_0^2 e^{-\alpha|x|} dx \right)$$

$$4\alpha \int_{-1}^0 e^{\alpha x} dx + 4\alpha \int_0^2 e^{-\alpha x} dx = 5$$

$$4\alpha \left(\frac{e^{\alpha x}}{\alpha} \right)_{-1}^0 + 4\alpha \left(\frac{e^{-\alpha x}}{-\alpha} \right)_0^2 = 5$$

$$4(1 - e^{-\alpha}) - 4(e^{-2\alpha} - 1) = 5$$

$$4e^{-2\alpha} + 4e^{-\alpha} - 3 = 0$$

Put $e^{-\alpha} = t$

$$4t^2 + 4t - 3 = 0$$

$$t = \frac{1}{2} \quad t = -\frac{3}{2} \text{ (Rejected)}$$

$$e^{-\alpha} = \frac{1}{2}$$

$$\Rightarrow \alpha = \ln 2$$

73. The value of the integral $\int_1^2 \left(\frac{t^4 + 1}{t^6 + 1} \right) dt$ is

समाकलन $\int_1^2 \left(\frac{t^4 + 1}{t^6 + 1} \right) dt$ का मान है :

(A) $\tan^{-1} \frac{1}{2} - \frac{1}{3} \tan^{-1} 8 + \frac{\pi}{3}$

(B) $\tan^{-1} \frac{1}{2} + \frac{1}{3} \tan^{-1} 8 - \frac{\pi}{3}$

(C) $\tan^{-1} 2 - \frac{1}{3} \tan^{-1} 8 + \frac{\pi}{3}$

(D) $\tan^{-1} 2 + \frac{1}{3} \tan^{-1} 8 - \frac{\pi}{3}$

Ans. D

Sol.
$$= \int_1^2 \frac{(t^4 + 1)(t^2 + 1)}{(t^6 + 1)(t^2 + 1)} dt$$

$$= \int_1^2 \frac{t^6 + t^4 + t^2 + 1}{(t^6 + 1)(t^2 + 1)} dt$$

$$= \int_1^2 \frac{t^6 + 1}{(t^6 + 1)(t^2 + 1)} dt + \int_1^2 \frac{t^4 + t^2}{(t^6 + 1)(t^2 + 1)} dt$$

$$= \int_1^2 \frac{1}{t^2 + 1} dt + \int_1^2 \frac{t^2}{t^6 + 1} dt,$$

$$I = \left[\tan^{-1} t \right]_1^2 + I_1$$

$$\tan^{-1} 2 - \frac{\pi}{4} + I_1 \quad \tan^{-1} 1 - \frac{\pi}{4} + I_1$$

$$I_1 = \int_1^2 \frac{t^2}{t^6 + 1} dt$$

put $t^3 = \theta \cdot 3t^2 dt = d\theta$

$t = 1$ then $\theta = 1$

$t = 2$ then $\theta = 8$

$$= \frac{1}{3} \int_1^8 \frac{d\theta}{\theta^2 + 1}$$

$$= \frac{1}{3} \left[\tan^{-1}(\theta) \right]_1^8$$

$$= \frac{1}{3} \left[\tan^{-1} 8 - \frac{\pi}{4} \right]$$

$$I = \tan^{-1} 2 - \tan^{-1} (1) + \frac{1}{3} \tan^{-1} 8 - \frac{\pi}{12}$$

$$\tan^{-1} 2 + \frac{1}{3} \tan^{-1} 8 - \frac{\pi}{3}$$

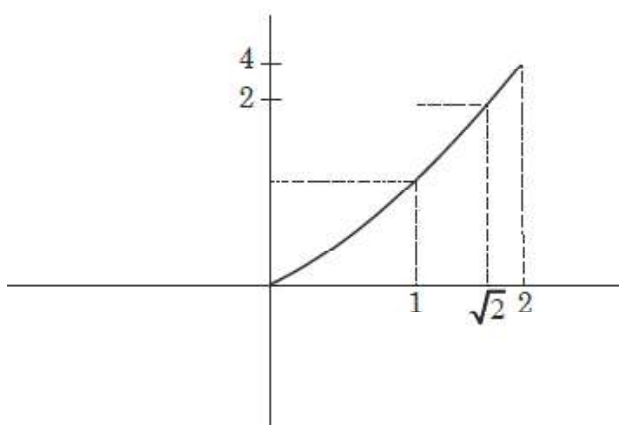
74. Let $[x]$ denote the greatest integer $\leq x$. Consider the function $f(x) = \max \{x^2, 1 + [x]\}$. Then the value of the integral $\int_0^2 f(x) dx$ is

माना $[x]$ महत्तम पूर्णांक $\leq x$ है। माना फलन $f(x) = \max \{x^2, 1 + [x]\}$ है। तब समाकलन $\int_0^2 f(x) dx$ का मान है :

- (A) $\frac{8+4\sqrt{2}}{3}$ (B) $\frac{5+4\sqrt{2}}{3}$ (C) $\frac{1+5\sqrt{2}}{3}$ (D) $\frac{4+5\sqrt{2}}{3}$

Ans. B

Sol.



$$A = \int_0^1 1 \cdot dx + \int_1^{\sqrt{2}} 2 dx + \int_{\sqrt{2}}^2 x^2 dx$$

$$= 1 + 2\sqrt{2} - 2 + \frac{8}{3} - \frac{2\sqrt{2}}{3}$$

$$\frac{5}{3} + \frac{4\sqrt{2}}{3},$$

75. If $I = \int_1^2 \frac{dx}{\sqrt{2x^3 - 9x^2 + 12x + 4}}$, then :

यदि $I = \int_1^2 \frac{dx}{\sqrt{2x^3 - 9x^2 + 12x + 4}}$ है, तब :

- (A) $\frac{1}{16} < I^2 < \frac{1}{9}$ (B) $\frac{1}{8} < I^2 < \frac{1}{4}$ (C) $\frac{1}{9} < I^2 < \frac{1}{8}$ (D) $\frac{1}{6} < I^2 < \frac{1}{2}$

Ans. C

Sol. $I = \int_1^2 \frac{dx}{\sqrt{2x^3 - 9x^2 + 12x + 4}}$

Let $f(x) = 2x^3 - 9x^2 + 12x + 4$

$f'(x) = 6x^2 - 18x + 12$

$f'(x) = 6(x-1)(x+2)$

$f'(x) < 0$ in $x \in (1, 2)$

$f(x) \downarrow$

$\frac{1}{\sqrt{f(x)}} \uparrow$

$(2-1) \times \frac{1}{\sqrt{f(1)}} < I < (2-1) \frac{1}{\sqrt{f(2)}}$

$\frac{1}{3} < I < \frac{1}{2\sqrt{2}}$

$\frac{1}{9} < I^2 < \frac{1}{8}$

76. Let $\alpha \in (0, \pi/2)$ be fixed. If the integral $\int \frac{\tan x + \tan \alpha}{\tan x - \tan \alpha} dx =$

$A(x) \cos 2\alpha + B(x) \sin 2\alpha + C$, where C is a constant of integration then the functions $A(x)$ and $B(x)$ are respectively

- (A) $x - \alpha$ and $\log_e |\cos(x - \alpha)|$ (B) $x + \alpha$ and $\log_e |\sin(x - \alpha)|$
(C) $x - \alpha$ and $\log_e |\sin(x - \alpha)|$ (D) $x + \alpha$ and $\log_e |\sin(x + \alpha)|$

माना $\alpha \in (0, \pi/2)$ दिया है। यदि समाकल $\int \frac{\tan x + \tan \alpha}{\tan x - \tan \alpha} dx =$

$A(x) \cos 2\alpha + B(x) \sin 2\alpha + C$, जहाँ C एक समाकलन अचर है, तो फलन $A(x)$ तथा $B(x)$ क्रमशः हैं:

- (A) $x - \alpha$ तथा $\log_e |\cos(x - \alpha)|$ (B) $x + \alpha$ तथा $\log_e |\sin(x - \alpha)|$
(C) $x - \alpha$ तथा $\log_e |\sin(x - \alpha)|$ (D) $x + \alpha$ तथा $\log_e |\sin(x + \alpha)|$

Ans. C

sol.
$$\int \frac{\tan x + \tan \alpha}{\tan x - \tan \alpha} dx$$

$$= \int \frac{\sin(x + \alpha)}{\tan(x - \alpha)} dx \quad \text{let } x - \alpha = t$$

$$\Rightarrow dx = dt$$

$$= \int \frac{\sin(t + 2\alpha)}{\sin t} dt$$

$$= [\cos 2\alpha + \sin 2\alpha \cdot \cot t] dt$$

$$= \cos 2\alpha \cdot t + \sin 2\alpha \cdot \ln |\sin t| + c$$

$$= (x - \alpha) \cdot \cos 2\alpha + \sin 2\alpha \cdot \ln |\sin(x - \alpha)| + c$$

77. Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} + \frac{5}{x(x^5 + 1)} y = \frac{(x^5 + 1)^2}{x^7}$, $x > 0$. If $y(1) = 2$, then $y(2)$ is equal to :

माना अवकल समीकरण $\frac{dy}{dx} + \frac{5}{x(x^5 + 1)} y = \frac{(x^5 + 1)^2}{x^7}$, $x > 0$ का हल $y = y(x)$ है। यदि $y(1) = 2$ है, तो $y(2)$ बराबर है :

- (A) $\frac{679}{128}$ (B) $\frac{637}{128}$ (C) $\frac{697}{128}$ (D) $\frac{693}{128}$

Ans. D

Sol. I.F = $e^{\int \frac{5dx}{x(x^5 + 1)}} = e^{\frac{5x^{-6}}{(x^5 + 1)}}$

Put, $1 + x^{-5} = t \Rightarrow -5x^{-6} dx = dt$

$$\Rightarrow e^{\int \frac{-dt}{t}} = \frac{1}{t} = \frac{x^5}{1 + x^5}$$

$$y \cdot \frac{x^5}{1 + x^5} = \int \frac{x^5}{(1 + x^5)} \times \frac{(1 + x^5)^2}{x^7} dx$$

$$= \int x^3 dx + \int x^{-2} dx$$

$$y \cdot \frac{x^5}{1 + x^5} = \frac{x^4}{4} - \frac{1}{x} + c$$

Given than : $x = 1 \Rightarrow y = 2$

$$2 \cdot \frac{1}{2} = \frac{1}{4} - 1 + c$$

$$c = \frac{7}{4}$$

$$y \cdot \frac{x^5}{1+x^5} = \frac{x^4}{4} - \frac{1}{x} + \frac{7}{4}$$

Now put, $x = 2$

$$y \cdot \left(\frac{32}{33} \right) = \frac{21}{4}$$

$$y = \frac{693}{128}$$

78. Let $y = y(x)$, $y > 0$, be a solution curve of the differential equation $(1+x^2)dy = y(x-y)dx$. If $y(0) = 1$ and $y(2\sqrt{2}) = \beta$, then :

माना अवकल समीकरण $(1+x^2)dy = y(x-y)dx$ का एक हल वक्र $y = y(x)$, $y > 0$ है। यदि $y(0) = 1$ तथा $y(2\sqrt{2}) = \beta$ हैं, तो :

(A) $e^{\beta-1} = e^{-2}(3+2\sqrt{2})$

(B) $e^{3\beta-1} = e(3+2\sqrt{2})$

(C) $e^{3\beta-1} = e(5+\sqrt{2})$

(D) $e^{\beta-1} = e^{-2}(5+\sqrt{2})$

Ans. B

Sol. $y(x-y)dx = (1+x^2)dy$

$$\frac{y(x-y)}{1+x^2} = \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{x}{1+x^2}y - \frac{y^2}{1+x^2}$$

$$\frac{dy}{dx} - \frac{x}{1+x^2}y = \frac{-y^2}{1+x^2}$$

$$\frac{-1}{y^2} \frac{dy}{dx} + \frac{x}{1+x^2} \frac{1}{y} = \frac{1}{1+x^2}$$

Let $\frac{1}{y} = t$

$$\therefore \frac{-1}{y^2} \frac{dy}{dx} = \frac{dt}{dx}$$

$$\therefore \frac{dt}{dx} + \frac{x}{1+x^2} t = \frac{1}{1+x^2} \text{ linear in } t$$

$$\text{I.F.} = e^{\int \frac{x}{1+x^2} dx} = e^{\frac{1}{2} \ln(1+x^2)} = \sqrt{1+x^2}$$

$\therefore$ solution is

$$t\sqrt{1+x^2} = \int \frac{\sqrt{1+x^2}}{1+x^2} dx + c$$

$$\frac{1}{y} \sqrt{1+x^2} = \int \frac{1}{\sqrt{1+x^2}} dx + c$$

$$\frac{1}{y} \sqrt{1+x^2} = \ln(x + \sqrt{x^2+1}) + c$$

$$y(0)=1 \Rightarrow 1=0+c \Rightarrow c=1$$

$$\text{So at } x=2\sqrt{2} \text{ we have } \frac{3}{\beta} = \ln(2\sqrt{2}+3)+1 \Rightarrow 3\beta^{-1} = \ln e(2\sqrt{2}+3)$$

$$\Rightarrow e^{3\beta^{-1}} = e(3+2\sqrt{2})$$

79. Let the solution curve $y = y(x)$ of the differential equation

$$\frac{dy}{dx} - \frac{3x^5 \tan^{-1}(x^3)}{(1+x^6)^{3/2}} y = 2x \exp \left\{ \frac{x^3 - \tan^{-1} x^3}{\sqrt{(1+x^6)}} \right\} \text{ pass through the origin. Then } y(1) \text{ is equal to :}$$

माना अवकल समीकरण $\frac{dy}{dx} - \frac{3x^5 \tan^{-1}(x^3)}{(1+x^6)^{3/2}} y = 2x \exp \left\{ \frac{x^3 - \tan^{-1} x^3}{\sqrt{(1+x^6)}} \right\}$ का हल वक्र $y = y(x)$ मूल बिंदु से होकर जाता है।

तो $y(1)$ बराबर है :

(A) $\exp\left(\frac{\pi-4}{4\sqrt{2}}\right)$ (B) $\exp\left(\frac{1-\pi}{4\sqrt{2}}\right)$ (C) $\exp\left(\frac{4+\pi}{4\sqrt{2}}\right)$ (D) $\exp\left(\frac{4-\pi}{4\sqrt{2}}\right)$

Ans. D

Sol. $\frac{dy}{dx} - \frac{3x^5 \tan^{-1} x^3}{(1+x^6)^{3/2}} y = 2x \exp \left\{ \frac{x^3 - \tan^{-1} x^3}{\sqrt{1+x^6}} \right\}$

$$\text{I.f.} = e^{-\int \frac{3x^5 \tan^{-1} x^3}{(1+x^6)^{3/2}} dx}$$

$$= e^{-\int \frac{3x^2 \cdot x^3 \tan^{-1} x^3}{(1+x^6)^{3/2}} dx}$$

Let $\tan^{-1} x^3 = t \Rightarrow \frac{3x^2}{1+x^6} dx = dt$

$$= e^{-\int \frac{x^3}{\sqrt{1+x^6}} \cdot \frac{3x^2 \tan^{-1} x^3}{(1+x^6)} dx}$$

$$= e^{-\int \frac{\tan t}{\sec t} \cdot t \cdot dt}$$

$$= e^{-\int t \sin t \cdot dt} = e^{(t \cos t - \sin t)}$$

$$= e^{\left(\frac{\tan^{-1} x^3}{\sqrt{1+x^6}} \cdot \frac{x^3}{\sqrt{1+x^6}} \right)}$$

Hence solutions of the differential equation is

$$y \cdot e^{\frac{\tan^{-1} x^3}{\sqrt{1+x^6}} - \frac{x^3}{\sqrt{1+x^6}}} = \int 2x \cdot e^{\frac{x^3 - \tan^{-1} x^3}{\sqrt{1+x^6}}} \cdot e^{\frac{\tan^{-1} x^3 - x^3}{\sqrt{1+x^6}}} \cdot dx + C$$

$$= x^2 + C$$

Passes through (0, 0) $\Rightarrow C = 0$

Hence $y(1) = e^{\frac{1}{\sqrt{2}} \left(1 - \frac{\pi}{4} \right)}$

80. If $\int \frac{\cos x \, dx}{\sin^3 x (1 + \sin^6 x)^{2/3}} = f(x)(1 + \sin^6 x)^{1/\lambda} + c$ where c is a constant of integration, then $\lambda f\left(\frac{\pi}{3}\right)$ is equal to :

यदि $\int \frac{\cos x \, dx}{\sin^3 x (1 + \sin^6 x)^{2/3}} = f(x)(1 + \sin^6 x)^{1/\lambda} + c$ है, जहाँ c एक समाकलन अचर है, तो $\lambda f\left(\frac{\pi}{3}\right)$ का मान है :

- (A) 2 (B) $-\frac{9}{8}$ (C) $\frac{9}{8}$ (D) -2

Ans. D

Sol. $\sin x = t$

$$\cos x \, dx = dt$$

$$I = \int \frac{dt}{t^3(1+t^6)^{\frac{2}{3}}}$$

$$I = \int \frac{t^{-7}}{(1+t^{-6})^{\frac{2}{3}}} dt$$

$$1 + t^{-6} = r^3$$

$$\frac{dt}{t^7} = -\frac{1}{2} r^2 dr$$

$$-\frac{1}{2} \int dr = -\frac{1}{2} r + c$$

$$= -\frac{1}{2} \left(\frac{1 + \sin^6 x}{\sin^6 x} \right)^{\frac{1}{3}} + c$$

$$= -\frac{1}{2 \sin^2 x} (1 + \sin^6 x)^{\frac{1}{3}} + c$$